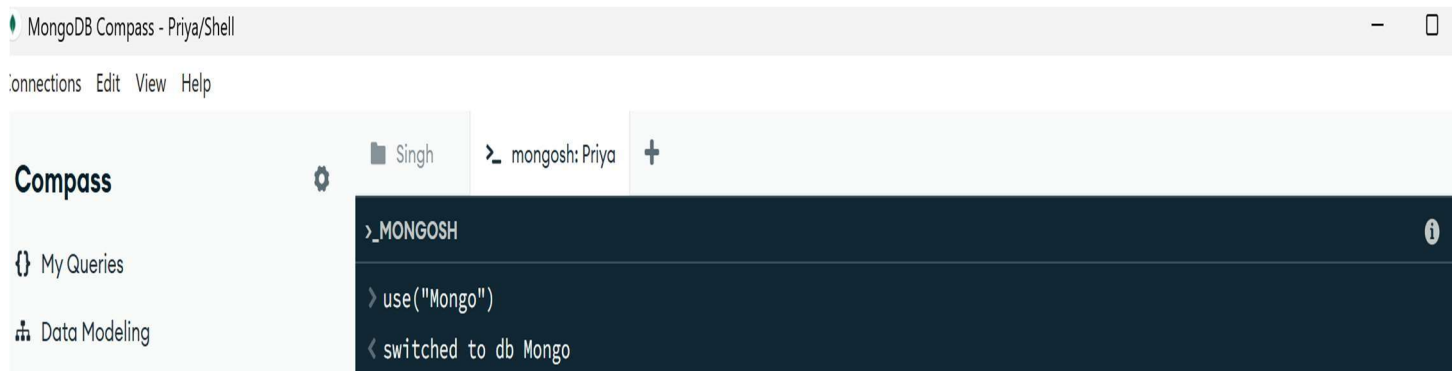


1. CREATE DATABASE

MongoDB Query:

```
use Students
```

Explanation: Creates or switches to the Student database.

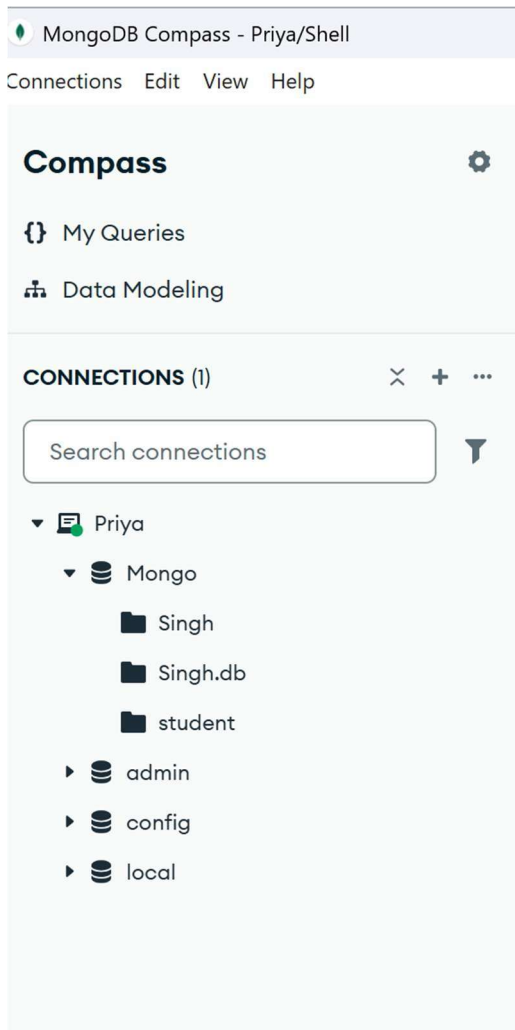


2. CREATE COLLECTION

MongoDB Query:

```
db.create Collection("students")
```

Explanation: Creates a collection named students.



3. INSERT 50 STUDENTS

MongoDB Query:

```
db.students.insertMany([  
  {roll:1, name:"Riya", age:18, marks:82, city:"Lucknow"},  
  {roll:2, name:"Kunal", age:19, marks:74, city:"Kanpur"},  
  {roll:3, name:"Sneha", age:20, marks:91, city:"Varanasi"},  
  {roll:4, name:"Raghav", age:18, marks:68, city:"Ayodhya"},  
  {roll:5, name:"Pooja", age:19, marks:85, city:"Jaunpur"},
```

{roll:6, name:"Arjun", age:20, marks:88, city:"Prayagraj"}, {roll:7, name:"Simran", age:18, marks:79, city:"Lucknow"},

{roll:8, name:"Mohit", age:19, marks:93, city:"Kanpur"},

{roll:9, name:"Kavya", age:20, marks:76, city:"Gorakhpur"},

{roll:10, name:"Ritesh", age:18, marks:64, city:"Bareilly"},

{roll:11, name:"Nikita", age:19, marks:87, city:"Jaunpur"},

{roll:12, name:"Sakshi", age:20, marks:95, city:"Lucknow"},

{roll:13, name:"Abhishek", age:18, marks:81, city:"Ayodhya"},

{roll:14, name:"Muskan", age:19, marks:89, city:"Prayagraj"},

{roll:15, name:"Rohit", age:20, marks:72, city:"Varanasi"},

{roll:16, name:"Komal", age:18, marks:86, city:"Lucknow"},

{roll:17, name:"Deepak", age:19, marks:78, city:"Kanpur"},

{roll:18, name:"Ishita", age:18, marks:57, city:"Gonda"},

{roll:19, name:"Manish", age:20, marks:92, city:"Lucknow"},

{roll:20, name:"Aarav", age:19, marks:66, city:"Bareilly"},

{roll:21, name:"Shivani", age:18, marks:83, city:"Lucknow"},

{roll:22, name:"Varun", age:19, marks:75, city:"Jaunpur"},

{roll:23, name:"Nandini", age:20, marks:80, city:"Kanpur"},

{roll:24, name:"Vikas", age:19, marks:88, city:"Ayodhya"},

{roll:25, name:"Payal", age:18, marks:69, city:"Prayagraj"},

{roll:26, name:"Rajat", age:20, marks:94, city:"Lucknow"},

{roll:27, name:"Isha", age:19, marks:84, city:"Varanasi"},

{roll:28, name:"Saurabh", age:18, marks:62, city:"Gorakhpur"},

{roll:29, name:"Tanvi", age:19, marks:77, city:"Kanpur"}, {roll:30, name:"Yash", age:20, marks:90, city:"Lucknow"},

{roll:31, name:"Preeti", age:18, marks:71, city:"Jaunpur"},

{roll:32, name:"Harshit", age:19, marks:89, city:"Lucknow"},

{roll:33, name:"Mansi", age:20, marks:59, city:"Ayodhya"},

{roll:34, name:"Dev", age:18, marks:85, city:"Kanpur"},

{roll:35, name:"Shubham", age:19, marks:96, city:"Prayagraj"},

{roll:36, name:"Anjali", age:20, marks:73, city:"Lucknow"},

{roll:37, name:"Nikhil", age:18, marks:82, city:"Varanasi"},

{roll:38, name:"Aditi", age:19, marks:67, city:"Gorakhpur"},

{roll:39, name:"Varsha", age:20, marks:93, city:"Lucknow"},

{roll:40, name:"Rishabh", age:19, marks:86, city:"Kanpur"},

{roll:41, name:"Kiran", age:18, marks:75, city:"Jaunpur"},

{roll:42, name:"Siddharth", age:20, marks:91, city:"Lucknow"},

{roll:43, name:"Pallavi", age:19, marks:63, city:"Ayodhya"},

{roll:44, name:"Aakash", age:18, marks:79, city:"Prayagraj"},

{roll:45, name:"Shalini", age:20, marks:87, city:"Lucknow"},

{roll:46, name:"Tarun", age:19, marks:70, city:"Kanpur"},

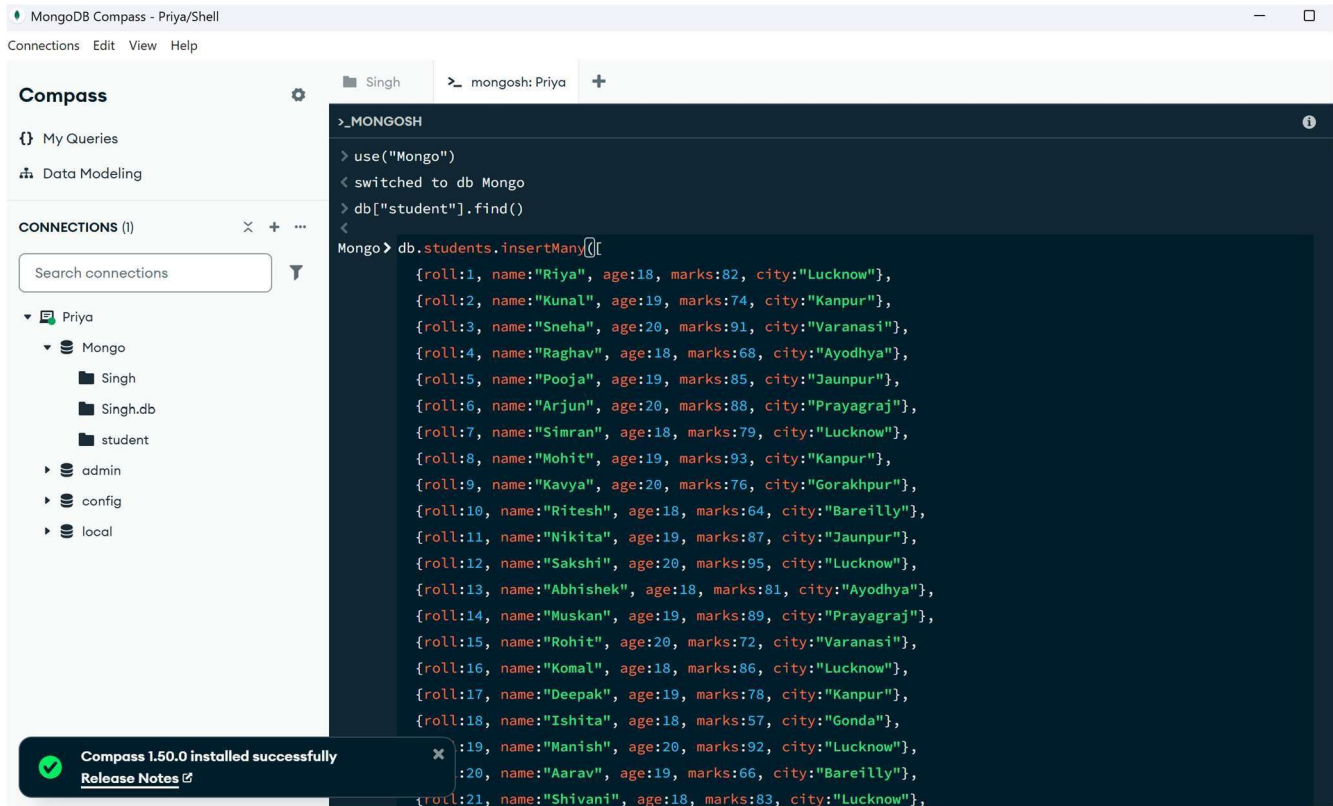
{roll:47, name:"Nisha", age:18, marks:61, city:"Gonda"},

{roll:48, name:"Viveka", age:20, marks:90, city:"Lucknow"},

{roll:49, name:"Sameer", age:19, marks:65, city:"Bareilly"},

{roll:50, name:"Divya", age:18, marks:92, city:"Varanasi"}

])



Explanation: Inserts multiple student documents into the collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db. students. Find ()
```

Explanation: Retrieves documents from the students collection.

```
>_MONGOSH
> db.students.find()
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf42'),
  roll: 1,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf43'),
  roll: 2,
  name: 'Kunal',
  age: 19,
  marks: 74,
  city: 'Kanpur'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf44'),
  roll: 3,
  name: 'Sneha',
  age: 20,
  marks: 91,
  city: 'Varanasi'
}
```

5. EQUAL TO — \$eq

MongoDB Query:

```
db. students. Find ({marks: {$eq: 88}})
```

Explanation: Finds students whose marks are exactly 88.

```
> db.students.find({marks: {$eq: 88}})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf47'),
  roll: 6,
  name: 'Arjun',
  age: 20,
  marks: 88,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf59'),
  roll: 24,
  name: 'Vikas',
  age: 19,
  marks: 88,
  city: 'Ayodhya'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.students.find({marks: {$gt: 85}})
```

Explanation: Finds students whose marks are greater than 85.

```
> db.students.find({marks: {$gt: 85}})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf44'),
  roll: 3,
  name: 'Sneha',
  age: 20,
  marks: 91,
  city: 'Varanasi'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf47'),
  roll: 6,
  name: 'Arjun',
  age: 20,
  marks: 88,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf49'),
  roll: 8,
  name: 'Mohit',
  age: 19,
  marks: 93,
  city: 'Kanpur'
}
```

7. LESS THAN — \$lt

MongoDB Query:

```
db.students.find({marks: {$lt: 60}})
```

Explanation: Finds students whose marks are less than 60.

```
> db.students.find({marks: {$lt: 60}})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf53'),
  roll: 18,
  name: 'Ishita',
  age: 18,
  marks: 57,
  city: 'Gonda'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf62'),
  roll: 33,
  name: 'Mansi',
  age: 20,
  marks: 59,
  city: 'Ayodhya'
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db. students. find({marks: {$gte: 90}})
```

Explanation: Finds students whose marks are 90 or above.

```
> db.students.find({marks: {$gte: 90}})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf44'),
  roll: 3,
  name: 'Sneha',
  age: 20,
  marks: 91,
  city: 'Varanasi'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf49'),
  roll: 8,
  name: 'Mohit',
  age: 19,
  marks: 93,
  city: 'Kanpur'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf4d'),
  roll: 12,
  name: 'Sakshi',
  age: 20,
  marks: 95,
  city: 'Lucknow'
}
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.students.find({marks: {$lte: 70}})
```

Explanation: Finds students whose marks are 70 or below.

```
> db.students.find({marks: {$lte: 70}})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf45'),
  roll: 4,
  name: 'Raghav',
  age: 18,
  marks: 68,
  city: 'Ayodhya'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf4b'),
  roll: 10,
  name: 'Ritesh',
  age: 18,
  marks: 64,
  city: 'Bareilly'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf53'),
  roll: 18,
  name: 'Ishita',
  age: 18,
  marks: 57,
  city: 'Gonda'
}
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(
  {roll:5},
  {$set:{marks:95}} )
```

Explanation: Updates the marks to 95 of the student with roll number 5.

```
> db.students.updateOne(
  {roll:5},
  {$set:{marks:95}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

11. DELETE ONE

OPERATION MongoDB Query:

```
db.students.deleteOne({roll:10})
```

Explanation: Deletes one document matching roll number 10.

```
> db.students.deleteOne({roll:10})
< {
  acknowledged: true,
  deletedCount: 1
}
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({  
  marks: {$lt:60} })
```

Explanation: Deletes all students having marks below 60.

```
> db.students.deleteMany({  
  marks: {$lt:60}  
})  
< {  
  acknowledged: true,  
  deletedCount: 2  
}
```

13. AND OPERATION — \$and

MongoDB Query:

```
db. students. find({  
  $and:[  
    {marks:{$gt:80}},  
    {age:19}  
  ]  
})
```

Explanation: Finds students satisfying both conditions.

```

> db. students. find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf46'),
  roll: 5,
  name: 'Pooja',
  age: 19,
  marks: 95,
  city: 'Jaunpur'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf49'),
  roll: 8,
  name: 'Mohit',
  age: 19,
  marks: 93,
  city: 'Kanpur'
}

```

14. OR OPERATION — \$or

MongoDB Query:

```

db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})

```

Explanation: Finds students from Lucknow or Kanpur.

```
> db.students.find({
  $or: [
    {city: "Lucknow"},
    {city: "Kanpur"}
  ]
})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf42'),
  roll: 1,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf43'),
  roll: 2,
  name: 'Kunal',
  age: 19,
  marks: 74,
  city: 'Kanpur'
}
```

15. SORT OPERATION

A.MongoDB Query:

```
db. students. find (). sort({marks : 1})
```

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

```
> db. students. find (). sort({marks : 1})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf70'),
  roll: 47,
  name: 'Nisha',
  age: 18,
  marks: 61,
  city: 'Gonda'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf5d'),
  roll: 28,
  name: 'Saurabh',
  age: 18,
  marks: 62,
  city: 'Gorakhpur'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf6c'),
  roll: 43,
  name: 'Pallavi',
  age: 19,
  marks: 63,
  city: 'Ayodhya'
}
```

B. MongoDB Query:

db. students. find (). Sort ({marks: -1})

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
> db.students.find().sort({marks: -1})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf64'),
  roll: 35,
  name: 'Shubham',
  age: 19,
  marks: 96,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf4d'),
  roll: 12,
  name: 'Sakshi',
  age: 20,
  marks: 95,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf46'),
  roll: 5,
  name: 'Pooja',
  age: 19,
  marks: 95,
  city: 'Jaunpur'
}
```

16. LIMIT OPERATION

MongoDB Query:

`db. students. find (). limit (5)`

Explanation: Displays only the first five documents.

```
> db.students.find().limit(5)
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf42'),
  roll: 1,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf43'),
  roll: 2,
  name: 'Kunal',
  age: 19,
  marks: 74,
  city: 'Kanpur'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf44'),
  roll: 3,
  name: 'Sneha',
  age: 20,
  marks: 91,
  city: 'Varanasi'
}
```

17. SEARCH BY CITY

MongoDB Query:

```
db.students.find({city:"Lucknow"})
```

Explanation: Finds all students whose city is Lucknow.

```
> db.students.find({city:"Lucknow"})
< {
  _id: ObjectId('6aba3857d3aca45bdc01bf42'),
  roll: 1,
  name: 'Riya',
  age: 18,
  marks: 82,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf48'),
  roll: 7,
  name: 'Simran',
  age: 18,
  marks: 79,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba3857d3aca45bdc01bf4d'),
  roll: 12,
  name: 'Sakshi',
  age: 20,
  marks: 95,
  city: 'Lucknow'
}
```

18. COUNT DOCUMENTS

MongoDB Query:

```
db. students. count Documents ()
```

Explanation: Counts documents currently present in the collection (performing after the delete function).

```
> db.students.countDocuments()
```

```
< 47
```